Design and Evaluation of a Low-Cost Educational Robotic Platform for Teleoperation and Teach-and-Repeat Manipulation

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***Abstract*— This paper presents the design and evaluation of a low-cost educational robotic platform for teleoperation and teach-and-repeat manipulation. The system consists of a 4-DOF robotic manipulator fabricated using 3D-printed PETG parts, six servomotors powered by an external 5 V, 5 A supply, an Arduino Uno board, and a PCA9685 PWM controller. During the teaching stage, the operator controls the robot joints through an analog joystick, while a Python application records the selected joint configurations. These configurations are subsequently reproduced automatically after verifying the predefined joint limits. MATLAB is used exclusively for offline three-dimensional visualization of the manipulator and the resulting end-effector motion. The experimental evaluation included 20 automatic replay trials and interaction tests with 20 users. The system achieved an 85% task success rate, a 95% joint-sequence replay consistency, and an average automatic cycle time of 47.6 s, compared with 145.8 s during manual teleoperation, representing a 67.4% reduction in task execution time. In addition, 80% of the participants successfully completed the manipulation task. The results indicate that the proposed platform provides an accessible and reproducible alternative for practical instruction in teleoperation, joint-space control, teach-and-repeat programming, and repetitive robotic manipulation without requiring high-cost equipment.**

Keywords— robotic manipulator; automatic trajectory reproduction; low-cost robotics; teleoperation; teach-and-repeat; joint-space control.

I. Introduction

Robotic manipulators are widely used in automated processes that require repetitive grasping, transportation, sorting, and positioning operations. Recent advances in industrial robotics have driven the development of more flexible and reconfigurable systems capable of adapting their motions to changes in the task and workspace [1]. In this context, joint coordination and appropriate trajectory generation are essential to maintain stable motion, avoid inefficient configurations, and reduce the risk of mechanical interference during manipulation [2]. Furthermore, synchronization among actuators becomes particularly important when a task is divided into sequential motions that must be executed in a predefined order [3], especially in low-cost platforms, where mechanical backlash, actuator variability, and limited sensing capabilities may affect motion continuity and repeatability [4].

Despite these advances, the acquisition, integration, and maintenance costs of industrial manipulators continue to limit their adoption in educational laboratories, small enterprises, and experimental applications. This situation has encouraged the development of robotic platforms based on additive manufacturing, microcontrollers, servomotors, and open-access software tools. Although these systems do not achieve the precision, payload capacity, or robustness of industrial robots, they provide a functional alternative for studying teleoperation, joint control, trajectory programming, and small-scale task automation. Furthermore, their modular structure facilitates mechanical modifications and the integration of new interaction interfaces without requiring specialized infrastructure.

Teleoperation enables the robot's motion to be controlled through commands generated directly by the operator. Current studies employ inertial measurement units, potentiometers, handheld devices, haptic interfaces, and asymmetric mapping methods to convert human actions into robotic movements [5], [6]. Low-cost master–slave architectures that combine multithreaded data acquisition and communication have also been proposed to improve signal responsiveness and stability [7]. In industrial education environments, gamepad-based control interfaces have been compared with kinesthetic programming. The results indicate that conventional controllers enable the generation of compact trajectories, although they may increase the time required to program a task [8]. In addition, vision-based systems can capture hand movements and convert them into commands for articulated manipulators during grasping and positioning tasks [9].

Other approaches incorporate haptic assistance, programming by demonstration, and learning algorithms. Force feedback can guide the operator along a desired trajectory and reduce deviations during teleoperation [10]. Likewise, teaching and variable-speed reproduction strategies enable the repetition of human demonstrations and the acquisition of additional data for imitation learning [11]. Visuo-haptic platforms capable of combining teleoperation and autonomous reproduction in force-sensitive tasks have also been proposed, achieving low communication latency and reduced positioning errors [12]. Similarly, reinforcement learning-based controllers have been investigated to improve trajectory tracking in multi-degree-of-freedom manipulators [13]. However, these solutions generally require haptic devices, tracking systems, industrial robots, training processes, or computational resources that may be

unnecessary for repetitive tasks performedin structured environments.

Existing teleoperation and programming-by-demonstration approaches use diverse interfaces, sensing systems, and control strategies, including IMUs, force feedback, gamepads, vision, haptics, imitation learning, and reinforcement learning [5]–[13]. However, many require specialized hardware or complex algorithms. In contrast, this work proposes a low-cost educational platform based on joystick teleoperation, configuration recording, joint-limit verification, and automatic playback, aiming to provide a simple and reproducible solution for teaching teleoperation and teach-and-repeat manipulation.

For basic manipulation applications, the teaching-by-demonstration and playback approach represents a simpler alternative. In this method, the operator manually guides the robot through the required configurations, records a sequence of relevant joint positions, and subsequently commands their automatic reproduction. Unlike fully offline trajectory programming, this procedure enables a task to be defined through direct demonstration. However, its implementation in low-cost manipulators requires preserving the order of the recorded configurations, verifying joint limits, coordinating the actuators, and ensuring communication among the user interface, the controller, and the actuation system.

In this work, the term teaching refers specifically to a teach-and-repeat programming-by-demonstration procedure rather than to machine learning or adaptive learning. The system records a finite sequence of joint configurations provided by the operator and subsequently reproduces them in the same order; therefore, no learning-based generalization is performed.

This work presents a low-cost teleoperation system with teaching-by-demonstration and playback for the automatic execution of trajectories in a four-degree-of-freedom robotic manipulator. The platform integrates a PETG 3D-printed structure, six servomotors, an Arduino Uno board, a PCA9685 PWM controller, a joystick-based teleoperation interface, and a Python application developed to transmit commands, record joint configurations, and automatically reproduce motion sequences. In addition, MATLAB is used for offline three-dimensional visualization of the manipulator and the trajectory followed by the end effector.

The problem addressed in this work is the limited accessibility of industrial robotic platforms for practical instruction in teleoperation and repetitive task programming. Commercial manipulators and advanced demonstration-based systems often require specialized hardware, sensing systems, or computational resources that may not be available in educational laboratories with limited budgets. Therefore, there is a need for simple and reproducible physical platforms that allow students to experimentally study the complete process of manual teaching, configuration recording, motion reproduction, and performance evaluation using accessible components.

The main contribution of this work is the development and experimental evaluation of an accessible teach-and-repeat robotic platform intended for practical robotics education and small-scale automation training. Rather than proposing a new control or machine-learning algorithm, the work focuses on the integration of low-cost hardware, joystick-based joint teleoperation, joint-limit verification, configuration recording, automatic playback, and offline three-dimensional visualization within a single reproducible platform. The system was physically implemented and evaluated through repeated automatic manipulation trials and interaction experiments with multiple users. This approach provides an affordable alternative for teaching fundamental concepts of joint-space control, teleoperation, programming by demonstration, and repetitive task automation in environments where industrial robotic manipulators may not be available.

II. System Design And Teaching–Playback Method

A. Hardware Architecture

The physical system consists of a four-degree-of-freedom (4-DOF) robotic manipulator with a gripper-type end effector for grasping, transportation, and positioning tasks involving lightweight objects. The prototype integrates Commercial Off-The-Shelf (COTS) components and 3D-printed PETG parts, achieving a modular, lightweight, and low-cost structure. Six hobby-grade servomotors provide joint actuation, offering the required torque and angular resolution for trajectory validation in educational and small-scale automation applications.

B. Control Architecture

The control architecture consists of a joystick, an Arduino Uno board, a PCA9685 PWM controller, six servomotors, a Python application, and a three-dimensional MATLAB environment for visualization. The mechanical structure was fabricated using 3D printing with PETG and incorporates a gripper that enables basic tasks such as grasping, transportation, and positioning of lightweight objects.

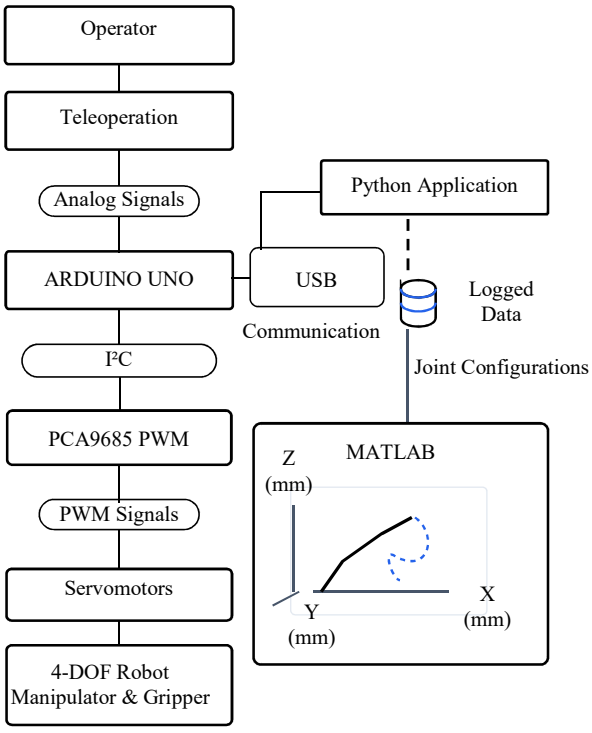


Fig. 1. Hardware and Software Architecture of the Teleoperation System with Teaching and Playback.

During teleoperation, the analog axes of the joystick define the motion references. The Arduino Uno identifies the joint to be actuated, determines the direction of motion, and updates its angular position accordingly. The reference commands are transmitted to the PCA9685 via I²C communication, which generates the PWM signals required to control the servomotors. The actuators are powered by an external regulated 5 V, 5 A power supply, while sharing a common ground with the Arduino Uno and the PWM controller. This configuration helps mitigate voltage drops and irregular motion when multiple joints operate simultaneously.

The Python application establishes communication with the Arduino Uno through a USB serial connection. Through this interface, commands are transmitted, joint configurations are stored, and the recorded positions are retrieved for subsequent automatic playback. MATLAB, in turn, is used exclusively for offline visualization of the manipulator and the end-effector trajectory; therefore, it does not participate in the physical control of the prototype.

C. Teleoperation and Configuration Recording

With this teleoperation method, each joint can be moved independently, without requiring the operator to enter Cartesian coordinates or angular values. The manipulator reaches the required configurations to perform the defined task through gradual joystick movements. Before applying each new reference command, the controller ensures that the requested position remains within the established limits of the corresponding joint.

The joint reference update can be represented as follows.

$$q_i(k+1) = q_i(k) + s_i \Delta q_i \tag{1}$$

where $q_i(k)$ is the current position of joint i, Δq_i is the angular increment defined in the program, and $s_i \in \{-1,0,1\}$ determines the direction of the displacement. The updated reference must satisfy.

$$q_{i,\min} \leq q_i(k+1) \leq q_{i,\max} \tag{2}$$

to avoid mechanical constraints, undesired forces, and contacts between the links.

During the teaching stage, the operator records only the relevant configurations of the task. The configuration stored at instant j is expressed as:

$$q^{(j)} = \begin{bmatrix} q_1^{(j)} & q_2^{(j)} & q_3^{(j)} & q_4^{(j)} \end{bmatrix}^T \tag{3}$$

and the taught trajectory is defined by the ordered sequence.

$$\mathcal{Q} = \left\{ q^{(1)}, q^{(2)}, \ldots, q^{(N)} \right\} \tag{4}$$

where N represents the total number of recorded configurations. Each point corresponds to an operation phase, such as approach, grasping, lifting, transportation, positioning, gripper opening, and return.

D. Automatic Playback and Visualization

After completing the teaching stage, the application retrieves the stored configurations and sequentially sends them to the Arduino Uno, which, before executing the motion, verifies again that the joint references remain within the defined limits. Once the values are confirmed as valid, they are transmitted to the PCA9685, which generates the PWM signals and commands the servomotors to reach the corresponding positions.

The playback process maintains the same order defined by the operator:

$$q^{(1)} \rightarrow q^{(2)} \rightarrow \cdots \rightarrow q^{(N)} \tag{5}$$

In the context of this work, task playback is based on an ordered sequence of taught joint configurations. The proposed implementation does not perform continuous trajectory interpolation, velocity or acceleration profile generation, Cartesian path optimization, or learning-based generalization between taught configurations. Each recorded configuration is treated as a discrete reference point, and the system reproduces these references sequentially under similar initial and workspace conditions.

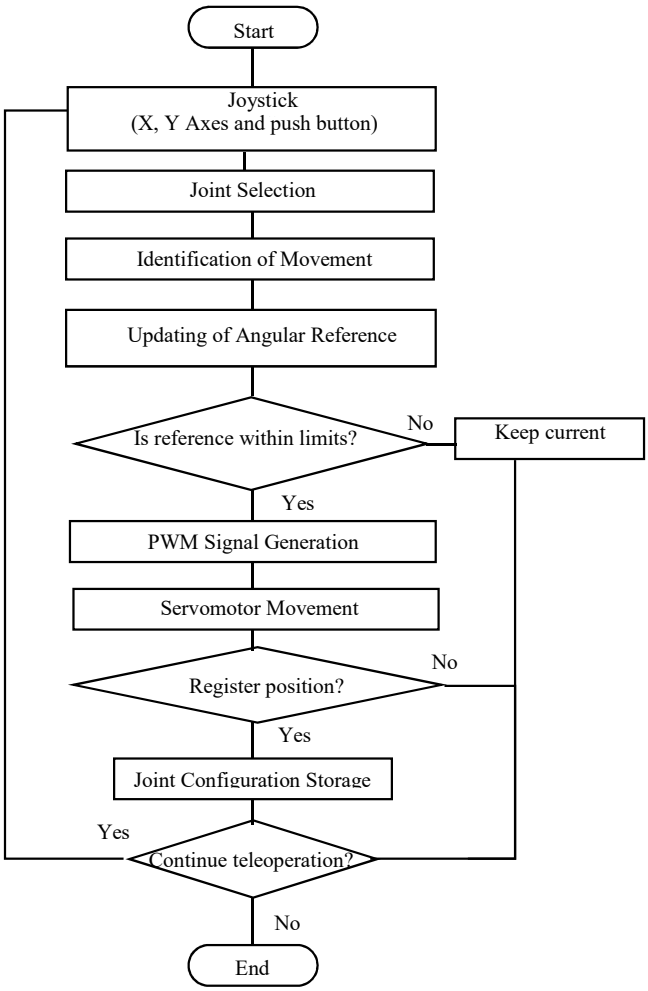


Fig. 2. Teaching and Playback Method Flow: Teleoperation, Limit Verification, Configuration Recording, Storage, and Automatic Playback.

With this procedure, the manipulator can repeat a previously taught task without continuous user intervention. The sequence can be executed multiple times, provided that the initial robot position, the location of the objects, and the workspace conditions remain unchanged.

As part of the validation process, the recorded configurations are represented in MATLAB using a simplified three-dimensional model. For each point (j), the position of the end effector is defined as:

$$p^{(j)} = [x^{(j)} \quad y^{(j)} \quad z^{(j)}]^T \tag{6}$$

and the Cartesian trajectory is obtained through the ordered set.

$$\mathcal{P} = \{p^{(1)}, p^{(2)}, \ldots, p^{(N)}\} \tag{7}$$

The consecutive connection of these points enables the visualization of the end-effector path and relates the joint variations to the spatial motion of the manipulator.

III. Experimental Setup And Results

A. Experimental Protocol and Metrics

The tests were conducted in a laboratory environment after completing the mechanical assembly, electronic integration, and initial calibration of the manipulator. The experimental setup included the four-degree-of-freedom robot (Fig.3), the teleoperation interface, a computer running the Python application, an Arduino Uno board, the PCA9685 PWM controller, and an external 5 V, 5 A power supply. A lightweight box was also used as a test object, and a designated area was defined for its placement.



Fig. 3. Low-Cost Four-Degree-of-Freedom Robotic Manipulator.

Before starting each trial, the robot was placed in a reference joint configuration. In addition, the power supply, serial communication, servomotor operation, wiring, and initial position of the object were verified. The task was performed in seven stages: approach, gripper closing, lifting, transportation, lowering, release, and return to the initial configuration.

The evaluation was conducted in two stages. First, a previously taught trajectory was executed automatically in 20 consecutive trials. Subsequently, 20 participants of different ages used the available commands to operate the manipulator manually. In both stages, an effort was made to keep the initial position of the box, the target area, and the robot's initial configuration unchanged.

Fig. 4 shows the three-dimensional model used to visually verify the trajectory traced by the end effector during automatic replay.

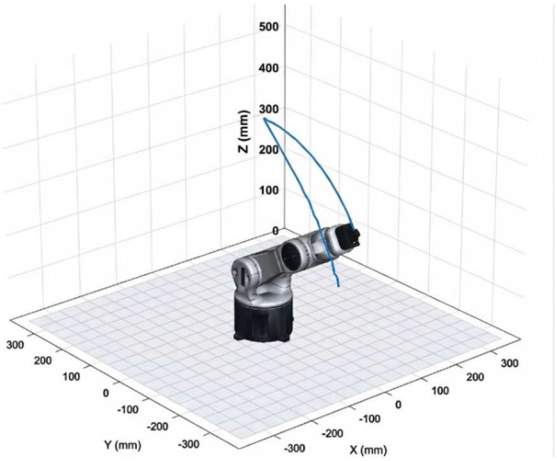


Fig. 4. Three-dimensional representation of the manipulator and end-effector trajectory in MATLAB.

Technical performance was evaluated using five indicators: task success rate, object drop rate, replay consistency, frequency of placement deviations, and mean cycle time. The success rate was defined as:

$$SR = \frac{N_s}{N_t} \times 100 \tag{8}$$

where N_s is the number of executions completed without object loss and N_t is the total number of trials. The drop rate was calculated as:

$$DR = \frac{N_d}{N_t} \times 100 \tag{9}$$

where N_d represents the number of trials in which the box became detached from the gripper during transport.

Replay consistency was estimated as:

$$CR = \frac{N_r}{N_t} \times 100 \tag{10}$$

where N_r represents the number of executions in which the manipulator replayed the stored joint sequence and reached a comparable final configuration. This metric was evaluated separately from the grasping outcome, since the trajectory can be replayed correctly even if the object slips from the gripper.

The mean automatic cycle time was obtained from

$$\bar{T}_a = \frac{1}{N_t} \sum_{j=1}^{N_t} T_{a,j} \tag{11}$$

whereas the relative reduction compared with manual teleoperation was calculated as:

$$TR = \frac{\bar{T}_m - \bar{T}_a}{\bar{T}_m} \times 100 \tag{12}$$

where $\bar{T}_m$ and $\bar{T}_a$ represent the mean manual operation and automatic replay times, respectively.



Fig. 5. Experimental Evaluation of the System.

B. Automatic Replay Performance

Of the 20 automatic executions performed, 17 successfully completed the grasping, transport, and release stages, representing a technical success rate of 85%. In the other three trials, the box became detached from the gripper during transport; therefore, the drop rate was 15%.

The joint sequence was replayed consistently in 19 of the 20 trials, corresponding to a consistency of 95%. This result shows that the system maintained the order of the recorded configurations and reached a comparable final configuration in almost all repetitions. The difference with respect to the success rate is explained by the fact that the grasping failures were mainly related to the interaction between the gripper and the object, rather than to incorrect trajectory replay.

TABLE I. QUANTITATIVE PERFORMANCE OF THE SYSTEM.

Metric	Result
Automatic trials	20
Successful executions	17
Technical success rate	85%
Object drops	3
Drop rate	15%
Consistent replays	19/20
Replay consistency	95%
Placements with observable deviation	02/20
Deviation frequency	10%
Observed deviations	7 and 10 mm
Maximum deviation	10 mm
Mean manual time	145.8 s
Mean automatic time	47.6 s
Time reduction	67.4%
Speedup factor	3.06

In two of the placements, deviations of approximately 7 and 10 mm from the target position were recorded. This represents a frequency of 10%, while the maximum displacement was 10 mm. Since the Cartesian position was not measured in all executions using the same metrological system, these data are reported as individual deviations rather than as a global mean positioning error.

The automatic cycle lasted an average of 47.6 s, a value obtained from the 20 replays. In contrast, manual operation required a mean time of 2.43 min (145.8 s). Therefore, the automatic mode reduced the mean task duration by 98.2 s, equivalent to 67.4%, and completed the execution approximately 3.06 times faster than manual teleoperation.

C. User Interaction Evaluation

The interaction evaluation involved 20 participants of different ages. Before the test, each participant received a brief explanation of the commands used to move the joints and control the gripper. The users then operated the manipulator to approach the box, grasp it, transport it, and place it in the designated area.

Of the 20 participants, 16 successfully completed the task, yielding a completion rate of 80%:

$$UTR = \frac{16}{20} \times 100 = 80\% \tag{13}$$

The other four participants experienced difficulties mainly when selecting or activating the commands. For this reason, these cases were considered user–interface interaction errors rather than solely mechanical, electronic, or communication failures.

TABLE II. USER INTERACTION RESULTS.

Result	Participants	Percentage
Task successfully completed	16	80%
Task not completed	4	20%
Total	20	100%

The perception survey assessed ease of learning, clarity of the controls, manipulator responsiveness, the relationship between commands and movements, joint smoothness, gripper positioning, the grasping mechanism, automatic replay, and overall performance. Most responses fell within the Excellent and Good categories. Fair ratings were mainly associated with the approach and precise positioning of the gripper.

D. Discussion

The results confirm the functional feasibility of the teach-and-repeat approach for performing basic manipulation tasks in structured environments. The technical success rate of 85% demonstrates that the system completed the operation in most trials, while the replay consistency of 95% indicates that the stored joint configurations were executed correctly.

The main quantitative advantage was observed in the cycle time. The 67.4% reduction compared with manual teleoperation indicates that storing and reusing a trajectory is beneficial when the same task must be repeated under similar conditions. This capability represents the main contribution of the proposed system.

The three object detachments suggest that the overall performance was constrained by the gripping capability of the gripper, the contact geometry, and the initial position of the box.

Therefore, increasing the gripping force or incorporating higher-friction contact surfaces could improve the success rate without modifying the replay method.

The deviations of 7 and 10 mm show that the platform exhibits placement variations attributable to mechanical backlash, the flexibility of the printed parts, the resolution of the servomotors, and the absence of external joint feedback. However, because the Cartesian error was not recorded in all 20 trials, it is not yet methodologically appropriate to establish a mean positioning accuracy for the manipulator.

Finally, the 80% completion rate obtained with users indicates that most participants were able to operate the system after a brief explanation. The errors observed in four participants show that the interface is a relevant component of overall performance and that it can be improved through labeled controls, visual feedback, and a guided operating sequence.

A limitation of the current implementation is that playback performance depends on maintaining conditions similar to those used during the teaching stage. Variations in the initial joint configuration, payload, object position or orientation, execution speed, or measurement noise were not systematically evaluated in this study. Therefore, the present platform should be interpreted as a teach-and-repeat educational system operating under structured conditions rather than as an adaptive trajectory-generation framework.

Overall, the results support the use of the prototype as an educational and small-scale automation platform. However, the lack of position sensors, force measurement, and complete statistical characterization limits its direct comparison with industrial manipulators or closed-loop control systems.

IV. CONCLUSIONS

A teach-and-repeat teleoperation system was designed and implemented for a low-cost, four-degree-of-freedom robotic manipulator. The platform enabled joint configurations to be recorded and grasping, transport, and release tasks to be replayed automatically.

In 20 trials, the system achieved a success rate of 85% and a replay consistency of 95%. The mean automatic execution time was 47.6 s, representing a 67.4% reduction compared with manual operation. In addition, 16 of the 20 users successfully completed the task.

The main failures were associated with object detachment, minor positioning deviations, and interface interaction errors. Future work will focus on improving the gripper, incorporating joint feedback, and optimizing the control interface.

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